

DEEP LEARNING FOR STRUCTURAL DESIGN

Stress Prediction and Optimisation in Cantilever Beam



Comparing a CNN and a U-Net for predicting maximum von Mises stress from strut-thickness distributions in a lattice cantilever beam.



Approach 1

CNN — direct max-stress prediction



Approach 2

U-Net — full stress-field prediction

APPROACH 1

Predicting Max Stress Directly with a CNN

A Convolutional Neural Network learns spatial patterns through convolution filters. Here it learns the relationship between strut-thickness distributions and the resulting maximum von Mises stress from FEA simulations.

- Each sample provides 226 thickness parameters and a von Mises stress field of ~3,305 values — reduced here to a single label: the sample's maximum stress.
- The 226-length thickness vector is reshaped into a 16×16 grid (with padding) so it can be fed through convolutional layers.
- Early stopping is added to help control overfitting.

Why it matters: a single scalar label discards where on the beam the stress is highest — the model never sees the spatial pattern.

Network Architecture

Input 16×16×1 grid

Conv2D (32) 3×3, ReLU

MaxPooling 2×2

Conv2D (64) 3×3, ReLU

MaxPooling 2×2

Flatten → vector

Dense (128) ReLU

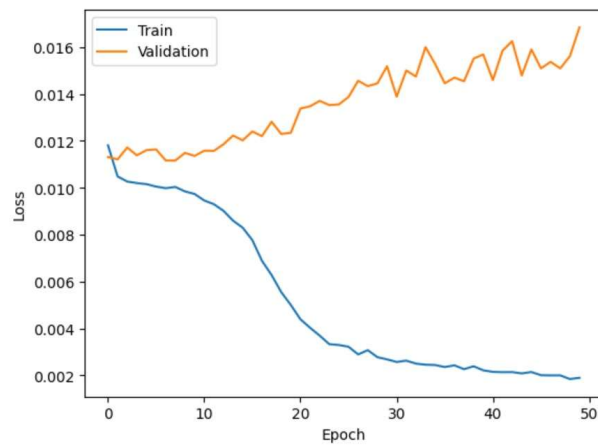
Dropout 0.3

Output 1 value — max stress

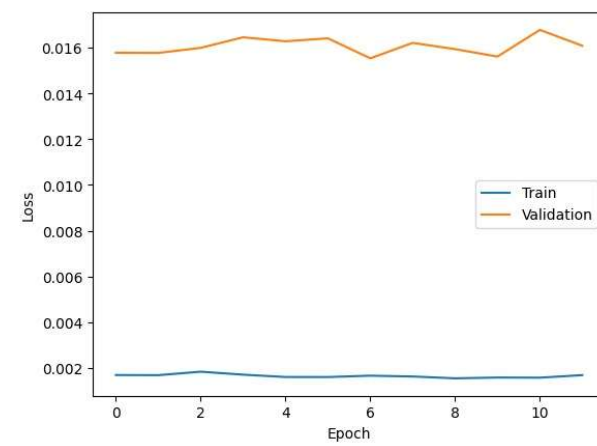
APPROACH 1 • RESULTS

Overfitting Shows Up Fast — and Early Stopping Only Helps So Much

50 Epochs — Before Early Stopping



Stopped at ~Epoch 10 — After Early Stopping



18,509,726

Predicted max stress

9,094,219

Actual max stress

Even after early stopping, a persistent gap between training and validation loss remains — evidence the model can't fully close, not just overfitting.

APPROACH 2

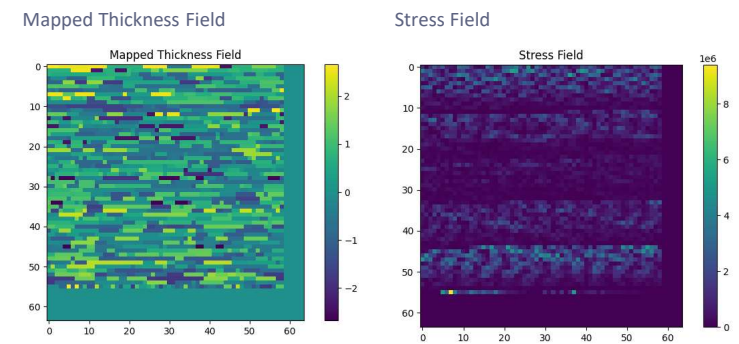
Predicting the Full Stress Field with a U-Net

- U-Net is a specialised CNN with a spatial, coordinate-based output — instead of one average max stress, it predicts the stress value at every node.
- An encoder → bottleneck → decoder structure is used, since the network needs both local detail and the global stress field.
- Built with Keras' Functional API, since the network branches and merges rather than stacking layers sequentially like the CNN.

Data Mapping



Network Input → Target



The thickness distribution (input) is mapped alongside its resulting stress field (target) so the U-Net learns the spatial relationship between the two directly.

APPROACH 2 · TRAINING

From Underfitting to a Working Field Predictor

- Conv2D layers learn spatial features — edges, local thickness gradients, local concentrations — from the mapped thickness grid.
- Max-pooling layers follow, within an encoder → bottleneck → decoder format.
- Goal: predict a full 64×64 grid of stress values from the 64×64 thickness input.

Initial model

~4,000 parameters, 50 epochs, batch 32

Result: underfitting

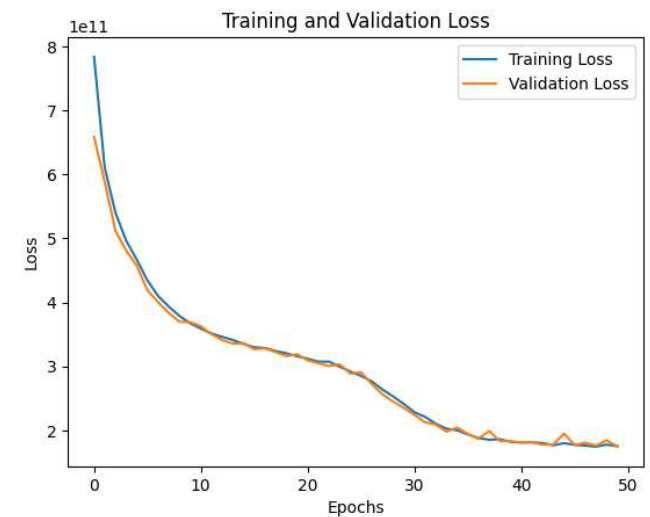


Scaled up

Capacity increased to 1,861,697 parameters

+ input data normalised

Training vs. Validation Loss (Final Model)



Loss falls smoothly and training/validation curves stay close together — a much better fit than the direct CNN.



CONCLUSION

The U-Net Wins by Keeping the Spatial Picture Intact

CNN — Direct Max-Stress Prediction

- Collapses the whole stress field to a single scalar — throws away spatial relationships between points
- A persistent train/validation gap remains even after early stopping
- Simpler and less data-hungry, but that simplicity caps its accuracy

U-Net — Full Stress-Field Prediction

- Predicts stress at every node, then takes the max — keeps spatial context intact
- Learns real spatial features: edges, thickness gradients, local concentrations
- Noticeably better prediction of maximum stress — at the cost of a larger, more data-hungry model

For this problem, predicting the full field first is the stronger approach — spatial context matters more than model simplicity.